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Detecting gravito-turbulence via non-thermal broadening in protostellar discs

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ABSTRACT

Chapter 1

Introduction

An accretion disc is an astrophysical system in which a central object is fed by the surrounding rotating material due to viscous forces. In particular, in this thesis, we will focus on discs around *Young Stellar Objects (YSO)*, such as a protostar. Such systems exhibit an extreme variability in their properties, which can also change a lot within different parts of the same disc.

The importance of these arrangements lies in the crucial role they play in the formation and accretion of the central object (stars or even supermassive black holes) and in planet formation, as we will see in Chapter 3. We will, in particular, enlighten the role of the discs in the redistribution of angular momentum, and try to identify the root cause of this process: as pointed out in chapter 2, a simple hydrodynamic description is insufficient to justify the magnitude and timescales of the angular momentum transport within a disc, so alternative ways are being sought.

A possible solution to the problem can come from the effects of the disc's self-gravity, in particular from gravitational instabilities which develop into characteristic spiral structures in the disc's density. In the evolution of unstable systems, will also emerge the importance of the cooling processes and the delicate balance between these two phenomenons.

PARTE SULLA PROGRAMMAZIONE E GLI SPETTRI, VEDDREMO

Chapter 2

Accretion discs dynamics

2.1 Basic equations

Accretion discs can usually be studied in the *thin disc approximation*, considering the thickness H of the disc to be much smaller than the radial distance from the central object R . In protostellar discs, for example, $H/R \approx 0.1$.

Treating the disc as infinitesimally thin, we will be allowed to integrate almost all the equations in the vertical direction, therefore dealing with quantities per unit surface or vertically averaged ones¹. For example, we will largely use the surface density $\Sigma = \int_{-\infty}^{\infty} \rho dz$. We will moreover consider the disc as vertically confined, neglecting the vertical component of the velocity.

A first approach in the description of these systems is employing the classical equations of hydrodynamics, as the dynamical active element is typically gaseous. Protostellar discs, for example, are made up of gas for about the 98

Such equations are the Navier-Stokes ones and the continuity equation:

$$\rho \left(\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla) \mathbf{v} \right) = -\nabla P + \nabla \cdot \boldsymbol{\sigma} - \rho \nabla \Phi , \quad (2.1)$$

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0 , \quad (2.2)$$

where ρ is the density, P is the pressure, Φ is the gravitational potential, and $\boldsymbol{\sigma}$ is the stress tensor describing viscosity.

Eq. 2.1 is simply an expression of Newton's II Law, as the left-hand side represents the acceleration, including the convection term due to velocity gradients, while on the right-hand side there are the different forces acting on the fluid.

¹Especially when dealing with these kind of quantities, particular caution must be taken to verify the applicability of the thin disc approximation

The nature of the stress tensor is, however, not known yet: a classical description in terms of shear viscosity does not, indeed, account for the timescale of the accretion, as we will see. This is the reason why this simple approach is insufficient, but it still allows us to derive some important properties of the disc.

These equations can be written using cylindrical polar coordinates due to the symmetry of the problem, so they can be broken down into the three components; each one can be studied in the limit cases in which gravity is dominated by the central object (*Keplerian rotation*) or by the disc's self-gravity (*self-gravitating disc*); in some cases, an analytical solution can be found.

For example, analysing the vertical component adopting the same approximations described in the next section for the radial equation, we can find an analytical solution for $\rho(z)$ in both cases² and define $H_{nsg} = \frac{c_s}{v_\phi} R$ and $H_{sg} = \frac{c_s^2}{\pi G \Sigma}$ as the typical scales for the disc thickness in the two regimes. Balancing the effects of the two gravitational potentials we find out that the vertical structure of the disc is affected by self-gravity even when its mass is much lower than the central object's mass, as we find that $\frac{M_D}{M} \approx \frac{H}{R}$, which is a small quantity in the thin disc approximation. In contrast, self-gravity has an impact on the radial structure of the disc only if the two masses are approximately the same, as the star gravity is much more important in this case.

Moreover, it's important to notice that, from the definition of H_{nsg} given above, a natural ordering between velocities emerges as an equivalent condition to the thin disc approximation, namely $c_s \ll v_\phi$.

In the next sections we will focus a bit more on the radial and angular equations. The first one is important to discuss the centrifugal equilibrium and derive the rotation curve of the disc ($v_\phi(R)$), which will play a crucial role in planet formation, as we will see in the next chapter (3.2.1); from the second one, instead, the equation regulating the transport of angular momentum and the relation $v_R \ll c_s$ can be derived.

2.1.1 Centrifugal balance

The radial component of 2.1 can be simplified assuming the disc is axisymmetric, so that the derivatives with respect to the azimuthal angle vanish³, assuming a classical shear viscosity and adopting the ordering of velocities shown above. We are

²see [Lodato, 2007] for a detailed derivation.

³In this derivation, as for Eq. 2.4 and 2.7 later, it is necessary to express correctly the derivative operators acting on vector in cylindrical coordinates; see [Adeeb, 2020] for the details.

therefore left with

$$\frac{v_\phi^2}{R} = \frac{1}{\rho} \frac{\partial P}{\partial R} + \frac{\partial \Phi}{\partial R}. \quad (2.3)$$

Considering a barotropic gas, $c_s^2 \doteq \frac{dP}{d\rho}$ (this comes straightforward from the wave equation for sound waves, which are classically understood as density or pressure waves in a fluid), so the first term on the right-hand side is in first approximation negligible and we can simply derive the rotation curve of the gas in the two limits introduced above: for a Keplerian system, the rotation curve is the classical $v_\phi(R) = \sqrt{\frac{GM}{R}}$, while for self-gravitating discs it will depend on the model; for example, in the Mestel model, $v_\phi = \sqrt{2\pi G \Sigma_0 R_0} = \text{const.}$

Since the total disc mass increases with the distance from the central body, realistic models generally include both the contributions from the star and the disc gravities, and a radius at which the transition between the two regimes occurs can be, therefore, determined, taking also into account the star accretion rate ($\dot{M}_*$) which reduces the importance of the disc over time. This transition affects the rotation curve, as well. It turns out that a very large accretion rate is needed for a disc to be significantly non-Keplerian at finite radii, which is typically not the case for protostellar discs.

Despite their low magnitude, pressure gradients can play a very important role when considering the interaction between the gas and solid components in the disc: the latter are not affected by pressure and will therefore describe precise Keplerian orbits (neglecting for the moment the disc self-gravity) with a slight different velocity with respect to the gas. This will lead to a radial motion of the solid particles, with important consequences described later in sect. 3.2.1.

Moreover, in the innermost and hottest parts of the disc, c_s can become very high, leading to sizable deviations from the Keplerian behaviour.

2.1.2 Viscosity and angular momentum transport

Let's start this analysis integrating Eq. 2.2 in the vertical direction; under the same assumptions as the previous section, we obtain

$$\frac{\partial \Sigma}{\partial t} + \frac{1}{R} \frac{\partial}{\partial R} (R \Sigma \tilde{v}_R) = 0 \quad (2.4)$$

where $\tilde{v}_R$ is the vertically averaged radial velocity⁴.

Considering now a stress tensor describing the classical shear viscosity, the only non-vanishing entry is $\sigma_{R\phi} = \eta R \frac{\partial \Omega}{\partial R}$ (η is the shear viscosity coefficient). We can

⁴In the following equations I'll omit the $\sim$ symbol.

then introduce the vertically averaged kinematic viscosity ν , defined as

$$\nu = \frac{1}{\Sigma} \int_{-\infty}^{\infty} \eta dz, \quad (2.5)$$

and integrate the ϕ component of Eq. 2.1 in the vertical direction (Later on, we'll denote with $T_{R\phi}$ the vertical integral of the stress tensor). The terms involving derivatives in the angular direction vanish (as in 2.4), so we are left with⁵

$$\Sigma \left(\frac{\partial v_\phi}{\partial t} + v_R \frac{\partial v_\phi}{\partial R} \right) = \frac{1}{R} \frac{\partial}{\partial R} \left(\nu \Sigma R^2 \frac{\partial \Omega}{\partial R} \right). \quad (2.6)$$

Combining this with Eq. 2.4, we recover the equation regulating angular momentum (per unit area) conservation, namely

$$\frac{\partial}{\partial t} (\Sigma R^2 \Omega) + \frac{1}{R} \frac{\partial}{\partial R} (\Sigma R^3 \Omega v_R) = \frac{1}{R} \frac{\partial}{\partial R} \left(\nu \Sigma R^3 \frac{\partial \Omega}{\partial R} \right), \quad (2.7)$$

where the second term on the left-hand side is the radial transport of angular momentum (negative if angular momentum is flowing in), and the right-hand side expresses the action of the torque $G(R) = 2\pi R^2 T_{R\phi}$ of viscous forces, which can also be seen as the divergence of the radial angular momentum flux. It is important to point out the dependency of this flux on the derivative of the angular velocity⁶, since it can justify the outward flow of angular momentum needed to allow accretion according to the process described in the next chapter.

Using the continuity equation again, we obtain the expression for the radial velocity and, substituting it back in Eq. 2.4 we find the master equation describing the evolution of accretion discs:

$$\frac{\partial \Sigma}{\partial t} = -\frac{1}{R} \frac{\partial}{\partial R} \left[\frac{1}{\frac{\partial}{\partial R} (R^2 \Omega)} \frac{\partial}{\partial R} \left(\nu \Sigma R^3 \frac{\partial \Omega}{\partial R} \right) \right]. \quad (2.8)$$

In the particular case of ν and Ω independent from Σ , this reduces to a simple diffusion equation for the surface density Σ , whose evolution is thus determined by viscosity, taking place on a timescale of the order of $t_v = R^2/\nu$. Assuming viscosity to be originated by collisions between gas molecules, we can derive an expression for ν and therefore compare this timescale to the one associated with the dynamics of the disc $t_d = \frac{T_{irr}}{2\pi} = \Omega^{-1}$.⁷ However, if we consider protostellar discs, this argument leads to timescales longer than the age of the Universe, as pointed out in [Lodato, 2007] (sect. 2.5).

⁵We assume the fields involved are sufficiently well-behaving so that $\int_{-\infty}^{\infty} \frac{\partial f}{\partial z} dz = 0$

⁶When trying to explain this phenomenon in kinetics terms, a paradox seem to arise, but see [Lodato, 2008] sect. 4.5.1 for a brief resolution.

⁷It's interesting to notice that this is also the typical timescale needed to reach hydrostatic equilibrium in the vertical direction for a Keplerian system. See [Lodato, 2008] sect. 6 for the details.

On the other hand, the ratio between the two timescales is equal to the Reynolds number $R_e = \frac{Rv_\phi}{\nu} = \frac{R^2\Omega}{\nu}$; since this number is generally very high, as just said, the fluid is in a turbulent condition. This suggests ascribing the transport of angular momentum to the mixing of fluid elements moving because of turbulence, rather than to classical viscosity. Unfortunately, we are not able to describe this process from first principles yet, but many hypotheses, including magnetohydrodynamic or gravitational instabilities acting like viscous forces, have been proposed.

2.2 The role of self-gravity

2.2.1 The α -prescription

Modern theories try to describe the transport of angular momentum due to instabilities in terms of an "effective viscosity" regulated by an unknown parameter α , which can be introduced to relate the vertically integrated pressure and stress tensor (whatever its real nature is)

$$T_{R\phi} = \frac{d \ln \Omega}{d \ln R} \alpha \Sigma c_s^2, \quad (2.9)$$

or equivalently ν and H in the case of classical viscosity ($\nu = \alpha c_s H$). From the latter relation, we can derive the constraint $\alpha < 1$, as the turbulence velocity is usually lower than the speed of sound. It is important to remark that this parameter is not necessarily a constant, but this constraint holds for every region in the disc. For protostellar discs, a typical value is $\alpha \approx 0.01$.

At thermal equilibrium, the heating rate due to this effective viscosity, which we will describe in sect. 2.3, must compensate the cooling rate of the gas; from this relation, we can derive an expression for α in the special case of an ideal gas:

$$\alpha = \left| \frac{d \ln \Omega}{d \ln R} \right|^{-2} \frac{1}{\gamma(\gamma - 1)\Omega\tau_{cool}}, \quad (2.10)$$

where γ is the ratio between the specific heats and τ_{cool} is the cooling timescale⁸.

Let's now focus on self-gravitating discs. To verify if this relation holds in observed systems, numerical simulations are used to compute the torque causing angular momentum transport and compare the relative value of α . In particular, assuming velocity and the self-gravitating potential (Φ) to be made up of a quasi-steady component ($\mathbf{v}, \Phi_0$) and a rapidly fluctuating one due to a non-necessarily axisymmetric

⁸Careful considerations on the cooling processes regulating the observed system are made to estimate this timescale.

perturbation⁹ ($\mathbf{u}, \Phi'$), with analogous calculations as for Eq. 2.7, one can derive the angular momentum conservation equation:

$$\frac{\partial}{\partial t} (\Sigma R^2 \Omega) + \frac{1}{R} \frac{\partial}{\partial R} (\Sigma R^3 \Omega v_R) = -\frac{1}{R} \frac{\partial}{\partial R} \left[R^2 \Sigma (\langle u_R u_\phi \rangle + \langle u_R^g e_\phi^g \rangle) \right]. \quad (2.11)$$

Here the angular brackets indicate azimuthal and vertical averages and $\mathbf{u}^g = \frac{-\nabla \Phi}{\sqrt{4\pi G \rho}}$. As in Eq. 2.7, the right-hand side represents a torque, which can be interpreted as due to an effective viscous force described by

$$\alpha_{sg} = \left| \frac{d \ln \Omega}{d \ln R} \right|^{-1} \frac{\langle u_R u_\phi \rangle + \langle u_R^g e_\phi^g \rangle}{c_s^2}. \quad (2.12)$$

The α value resulting from this equation and the expected one from Eq. 2.10 (or, equivalently, the associated torques) exhibit a relatively good agreement.

2.2.2 Gravitational instabilities and self-regulation

Self-gravity is responsible for the development of gravitational instabilities in discs, which result in characteristic spiral structures in the disc density. Having established that classical viscosity has a very small effect on the disc dynamics, we can go back to Eq. 2.1 and neglect the viscosity term; we can therefore carry out a linear local analysis of the behaviour of density waves in the case of tightly wound perturbation (WKB approximation).

To describe these waves, we introduce the radial wave-number k , the azimuthal wave-number m , and the pitch angle ψ ; in the WKB approximation, $\tan \psi = \frac{m}{kR} \ll 1$ and the dispersion relation is

$$(\omega - m\Omega)^2 = c_s^2 k^2 - 2\pi G \Sigma |k| + \kappa^2, \quad (2.13)$$

where ω is the frequency of the perturbation and κ is the epicyclic frequency, given by

$$\kappa^2 = \frac{2\Omega}{R} \frac{d(\Omega R^2)}{dR}. \quad (2.14)$$

The left-hand side represents the Doppler-shifted frequency, which results from a balance between pressure, self-gravity, and rotation. Self-gravity is the term that makes the system unstable, as it is negative: if it dominates, the frequency acquires an imaginary part which results in an exponentially growing perturbation; pressure

⁹Note that we're not considering perturbations due to gravitational instabilities yet, as the gravitational potential can be affected by any variation of the mass distribution. We're only analysing the effects of a perturbation on the disc self-gravity.

is effective in stabilizing small wavelength perturbations, while rotation dominates in the opposite regime.

From the simple solution of the inequality $(\omega - m\Omega)^2 > 0$, we thus find out that the system is stable if $Q = \frac{c_s \kappa}{\pi G \Sigma} > 1$. Since κ is usually of the same order as Ω , we expect the disc to be in a condition of marginal stability when the effect of the disc's self-gravity starts affecting the vertical structure of the disc¹⁰.

Moreover, Q is proportional to c_s , which is proportional to the temperature of the gas; therefore, if no other sources of heat are present, the disc cools down and becomes more unstable. Instabilities induce compression and shocks which dissipate energy, so the disc heats up again and goes back to a stable state; this way, the instabilities self-regulate, maintaining the disc in a condition of marginal stability, with Q fluctuating around 1.

For this mechanism to be effective, the cooling timescale needs to be not too small; otherwise, the heat produced by the instabilities, which evolve on the disc's dynamics timescale (Ω^{-1}), will be removed too rapidly and the disc will fragment. Numerical simulations reveal that proper values are $\tau_{cool} \approx 3-6 \Omega^{-1}$. Using Eq. 2.12, we can relate this critic to a maximum value for α_{sg} at which the disc will fragment, and interpret this threshold as an upper bound for the angular momentum that can be transported by self-gravity. In particular, this relation is derived in the case of a quasi-Keplerian disc¹¹ in a steady state.

Numerical simulations carried out using different models and values for γ return a threshold value of 0.06.

The most unstable wavelength is the value that minimizes the right-hand side term, $k_0 = \frac{\pi G \Sigma}{c_s^2} = \frac{1}{H_{sg}}$; from this relation, we can justify the application of the WKB approximation to thin discs, as $kR > \frac{R}{H_{sg}} \gg 1$, therefore $\tan \psi \ll 1$. It is important to note that, as the wavelength of the perturbations is linked with the disc's thickness, more massive discs tend to be more unstable for perturbations with small m .

It's worth emphasizing the fact that this is a local analysis of the problem, as α is introduced to relate the stress tensor to the local pressure. Simulations indicate that this approach is justified only if $H/R \approx M_D/M \lesssim 0.1$, thus light and massive discs appear to have different behaviours. In particular, this analysis can't be applied to massive discs, while for light discs both local and global effects can emerge, as the majority of the modes excited by the cooling process generally interact with one another and dissipate on short length scales, but some can last longer (especially the ones with higher wavelengths), propagating to longer distances.

The difference between light and massive discs emerges when studying the en-

¹⁰From the analysis of the vertical component of the Navier-Stokes equations, we can derive the ratio between the typical thickness for self and non-self-gravitating disc, namely $\frac{H_{sg}}{H_{nsg}} = \frac{c_s \Omega}{\pi G \Sigma}$

¹¹As pointed out at the beginning of sect. 2.1, this is the most common case for protostellar discs.

ergy flux associated with this angular momentum transport as well: extra terms arise in the relation between the flux and the stress tensor, which are therefore no longer proportional except in the special case of co-rotation (the wave pattern rotates at the same speed as the disc). Simulations point out that, even in this case, for massive discs global effects are non-negligible, leading to the emergence of several fleeting fierce spiral events with much larger α values, up to 0.5, determining a fast redistribution of the gas within the disc and thus a "global" angular momentum and energy transport.

2.3 Spectral energy distribution (SED)

As the gas falls into the potential well of the central object and accretes, it releases its gravitational potential energy, which is converted into heat mainly by the same process akin to viscosity responsible for angular momentum transport. Such a dissipation can therefore be expressed in terms of the stress tensor of Eq. 2.7 or 2.11, as the power dissipated in a rotational motion is given by the product of the torque and the angular velocity: dividing the disc into infinitesimal annuli of radial extension ΔR , the net torque exerted on an annulus at radius R by the neighbouring ones is simply $\frac{\partial G(R)}{\partial R} \Delta R$. Exploiting Leibniz's rule and considering that $\frac{\partial(G\Omega)}{\partial R}$ gives no contribution to the local heating of the disc as it is responsible for the transport of energy towards its outer edge, the power dissipated per unit area can be written as

$$D(R) = \frac{1}{2\pi R \Delta R} \frac{\partial \Omega}{\partial R} G \Delta R = R T_{R\phi} \frac{\partial \Omega}{\partial R} = \nu \Sigma \left(R \frac{\partial \Omega}{\partial R} \right)^2, \quad (2.15)$$

where the last equality holds for the particular case of shear viscosity.

In first approximation, this dissipation can be treated as blackbody radiation; from the Stefan-Boltzmann law, we can therefore derive the temperature of the disc, finding that it depends on the distance from the central object:

$$T_{\text{eff}}(R) = \left(\frac{1}{2\sigma_B} R \frac{\partial \Omega}{\partial R} T_{R\phi} \right)^{1/4}. \quad (2.16)$$

As a consequence, the full SED is the superposition of different blackbodies, a "stretched blackbody" spectrum.

In some cases, however, this simple argument does not hold, and a detailed analysis including the optical properties of the gas and dust should be carried out. In particular, when dealing with self-gravitating discs, the rotation curve becomes flatter and the self-regulation mechanism described in sect.2.2.2 results in higher temperatures at large radii, where the disc self-gravity effectively overcomes the central object's one. As a result, the SED of these discs is usually flatter and displays a secondary bump at large wavelengths.

Moreover, up to now we've ignored the heating due to the radiation coming from the star, which is often non-negligible in reality. Generally, both the "active" and the "passive" heating are present in a disc, and their relative importance is regulated by the comparison between the star accretion rate and the stellar luminosity.

In any case, the temperatures associated with the disc are much lower than a star photosphere one, so the full spectrum of a star-disc system will show an excess at near and mid-infrared wavelengths when compared to a single star SED; this characteristic is exploited both to spot and to classify these systems, as we will see in the next chapter.

Chapter 3

Protostellar discs

3.1 Formation, classification and observations

Stars develop in the core of turbulent molecular clouds; this core usually carries a specific angular momentum $j = 10^{21} \frac{\text{cm}^2}{\text{s}}$, which can be estimated from its *jeans length*, the linear scale of a system in which gravity is balanced by pressure¹, and the measurements of the angular velocity of the gas in the core. This value is several orders of magnitude larger than the typical value for a star².

To form a star³, the gas is therefore forced to lose angular momentum, and a disc-shaped object is the natural outcome of this process: instead of freely collapsing towards the centre, the gas can at most reach the minimum energy state of a Keplerian circular orbit, forming a ring of radius $R_0 = \frac{j^2}{GM}$ ($\approx 10^2$ - 10^4 AU for a star of 1 solar mass). The evolution of this gas ring is fully determined by Eq. 2.8, and it can be shown that it consists in a simultaneous inward and outward spreading (at least in the case of constant ν - see [Lodato, 2007], sect. 2.6), resulting in the formation of a disc. This disc is, in first approximation, still Keplerian, so the angular velocity decreases as a function of the radius, leading to the desired rearrangement of angular momentum, as it will flow outwards according to Eq. 2.7, allowing at the same time the gas to move inwards and feed the central protostar (main infall phase). In these first stages, the disc will be still wrapped in a gas cloud that will continuously collapse onto the initial ring, accreting the disc mass and partially compensating for the losses due to the infall process.

The infall is generally more rapid than the disc's accretion, so the importance of the disc self-gravity will gradually fade away, and the system will switch to a state

¹See [Lodato, 2008] for the explicit derivation.

²Our sun has a specific angular momentum of $10^{16} \frac{\text{cm}^2}{\text{s}}$.

³We are ignoring the effects of interactions between forming stars that could lead to a binary system and also magnetic field effects, but the process described is valid, anyway.

dominated by the star's gravity. As explained in the previous chapter, such systems show hardly mass-dependent properties, and are hence grouped on the basis of the stages of the process described above. In particular, the fundamental parameter to differentiate the classes described below is their SED, which exhibits an excess at longer wavelengths (with respect to a standard stellar photosphere) whose magnitude decreases as the system evolves, both because the gas envelope progressively disappears and the effective emission of the disc and the star becomes visible, and because the system becomes less gravitationally unstable.

3.1.1 Class 0 and I systems

These are the youngest objects, still enveloped by the gas of the original cloud progressively accreting the disc, which is, in turn, extremely massive. These systems have been associated with the observations of very massive stars ($M_* > 8M_\odot$), which enter the principal sequence while still accreting due to their large mass.

The presence of the gas envelope makes direct observations difficult, so there remains substantial uncertainty about the presence and the characteristics of the disc in the most massive and wide systems (O-type protostars, with masses over $20M_\odot$). Smaller systems (up to an extension of 10^3 AU), where the central object is a B-type star ($8-20M_\odot$), are instead recognised as proper discs and some more precise measurements can be performed. For example, an important parameter is the outflow rate from the inner parts of the disc (the star's accretion rate); its value is of the order of $10^{-4}-10^{-3}M_\odot/\text{yr}$, and it can be theoretically justified considering global angular momentum transport due to frequent brief spiral structures that originate in massive discs and are related to large α values, as described in sect. 2.2.2.

Abrupt increases in the accretion rate are identified as the cause of great outbursts, with a corresponding enhancement in luminosity by a few orders of magnitude, observed in a group of stars labelled as *FU Orionis objects*. These objects are understood to be quiescent Class I systems showing transient intense activity, and data from the observations are compatible with the hypothesis of self-gravitating discs with a mass of the same order of magnitude as the central object's one. The variation of the accretion rate could therefore be imputed to the development of gravitational instabilities, as suggested by numerical simulations as well, but a comparison with data from observations still lacks.

The more accepted explanation is, in fact, given in terms of thermal-viscous ("limit cycle") instabilities: in the general case of ν dependent on Σ , Eq. 2.8 becomes non-linear but some properties can still be deduced by rewriting it in terms of $\mu = \nu\Sigma$ which is essentially the mass flowing out of a disc annulus at radius R (as all the quantities are functions of the radius). If $\frac{\partial \mu}{\partial \Sigma} < 0$, indeed, the equation is no

longer a simple diffusion equation, as a decrease in the surface density is followed by an increase of the flux, resulting in a positive feedback mechanism for which more mass will flow away from the annulus. Solving the equation for the whole disc shows that when an instability is triggered at a certain radius as described above, a fast perturbation front will propagate inwards, ionizing the gas thus increasing the optical and infrared energy flux emitted by the disc, while a slow wavefront will move outwards until it reaches a zone where the surface density can be stable. The gas in the inner parts of the disc will be, therefore, consumed quickly (resulting in the observed outbursts), and will be slowly replaced by the outer materials, so the whole process will repeat after a long quiescent phase.

This mechanism is, however, not able to fully accomplish the observations (See [Lodato, 2008] sect. 7.1 for a detailed summary of the problems); this is why alternative explanations such as gravitational instability have been proposed.

3.1.2 Class II and III systems

Class II objects are also referred to as *Classical T-Tauri stars*. In these systems, the main infall phase has ended and the disc is visible. They are expected to be less extended than the previous ones and are generally identified with solar mass young stars. The observed systems attributed to this class can be both gravitationally stable and unstable, depending on the relation between the ratios M_D/M_* and H/R , where H can be estimated from the temperature of the disc, R from direct observations, M_* from the star luminosity, and M_D is obtained rescaling the mass derived from dust submm emission, in the assumption of an optically thin medium and provided the opacity of the dust is known.

The last measurement, especially, is subject to systematic errors and caveats regarding the applicability of the approximations employed, as the opacity and the dust size distribution can be very different from the interstellar medium ones, due to the presence of large dust grains (as we will see in sect. 3.2.1); therefore it is likely for the disc mass to be underestimated and for the system to be consequently gravitationally unstable.

For these objects, the accretion rate onto the star can be determined from the emission lines of the gas which falls into the star following the stellar magnetic field lines (*magneto-sphere attraction*), or from variations of intensity in the photospheric spectrum attributed to accretion shocks on the star's surface.

Class III objects are the most evolved ones: the gas has almost completely vanished and what remains is a rocky "debris disc". This stage is typically reached after a few million years, lifetime which is estimated from the fraction of stars which display an excess in infrared emission in young stellar clusters of different ages.

3.2 Planet formation

Protostellar discs are of particular interest because they are the sites where planets assemble. More than for specific significances of the discs themselves, indeed, they are generally studied in order to better understand this process and the origin of our own Solar System.

Up to now, thousands of extrasolar planets have been found, with large variability in terms of mass, eccentricity, and other parameters; most of the planets are found very close to the star they orbit around (less than 1 AU) but, actually, this is in large part a consequence of the intrinsic limitations of the detection methods deployed: the majority of them have been detected through the *transit* or the *radial velocity* methods⁴, which require being able to observe several orbits of the planet around the star to point out a periodic pattern and extract the planet's features. As a result, only the closest planets, which have short orbital periods, are typically discovered. Many of these planets are also too massive for having formed at such a distance, as the locally available mass given the disc's density is too small, suggesting that radial large-scale migrations can occur.

Trying to describe planet formation, two models have been developed; in both of them, the disc's self-gravity may play a crucial role.

3.2.1 The core accretion model

This model is able to explain both the formation of Earth-like rocky planets and gas giants (data from Jupiter and Saturn observations suggest that these planets have a solid core, too) in terms of the gradual gathering of solid elements of the disc forming dust grains of various dimensions thanks to inelastic scattering first and gravitational focusing later, when the core becomes sufficiently large. If the core is very massive, it can eventually attract a great amount of gas too, becoming a gas giant.

However, there's an intermediate phase in which the grains, which are now called planetesimals, are too big for inelastic scattering to be effective but too small to generate a significant gravitational potential. Therefore, a different explanation needs to be found. Moreover, the timescale associated with this process is generally too long.

In this context, gravitational instabilities due to the disc's self-gravity become particularly intriguing, as spiral structures determine the arising of local maxima in the pressure (seen as a function of the radius). As a consequence of fluidodynamics equations shown in the previous chapter, pressure gradients modify the rotation curve of the gas, but do not affect the solid component's motion, which will maintain

⁴Other classical methods used to spot exoplanets are *microlensing*, *direct imaging*, and *astrometry*. See [?] for the details about each method.

a Keplerian orbit and will therefore rotate more or less rapidly than the gas, depending on the sign of the gradient. Between the gas and the grains, a drag force will in turn develop, generating a radial motion of the solid chunks and concentrating them to the pressure maximum. For intermediate size planetesimals, in particular, the movement induced by this force occurs on a very short timescale and is very effective, as the solid component is neither too light (in this case it would be simply carried away by the gas), nor too heavy (the radial velocity would be negligible).

This mechanism could, in principle, solve the problems pointed out above, and numerical simulations are being built to verify it.

3.2.2 The gravitational instability model

As exposed in sect. 2.2.2, if the cooling rate of the disc is too high, the self-regulation mechanism is not effective, and the disc will rapidly fragment into gravitationally bound clumps, leading to the formation of gas giant planets without a solid core.

Considering that the most unstable wavelength attributed to the instabilities is $\lambda = 2\pi H$ (cfr. sect. 2.2.2), the mass of the fragment can be roughly estimated as $M_{frag} = \Sigma \lambda^2 = \frac{4\pi}{Q} \left(\frac{H}{R}\right)^3 M_*$. For example, for a typical unstable disc, with $Q \approx 1$ and $H/R \approx 0.1$, around a star like our sun, $M_{frag} \approx 12M_{Jupiter}$.

It is therefore evident that this model is incapable of accounting for the existence of rocky planets, and can only explain the formation of the most massive planets or brown dwarfs. Moreover, the cooling rate is generally large enough to induce fragmentation only at considerable distances from the star (50-100 AU). Confirmations to this model are thus still lacking because, as said in the previous section, it is quite difficult to spot planets very far from their host star.

In the light of what I've exposed above, the possibility of detecting gravito-turbulences by analysing the spectrum of these discs could be of great help to confirm the role of gravitational instabilities in planet formation and validate both these models.

3.3 Non-thermal broadening

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Ringraziamenti